SAS: Scene-Annotated AudioSet Strong
for Joint Scene-Event Analysis

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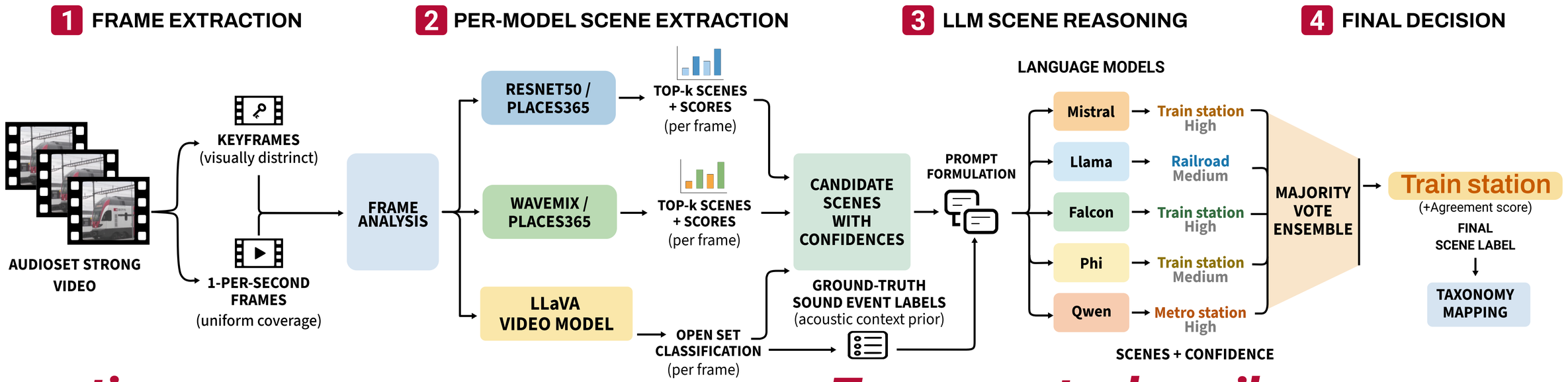
MOTIVATION

Same sound, different meaning: a *siren wails* on a *street* but *jars* in a *library*.
Context-aware listening needs both **events** and **scenes.**
The gap: AudioSet Strong has **events but no scenes**, so what characterizes a scene is **untested at scale**.

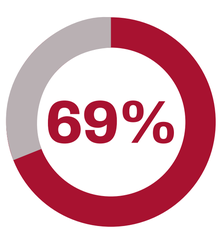
CONTRIBUTIONS

1. **Annotations**. Scene labels under two taxonomies for AudioSet Strong.
2. **Analysis**. How events distinguish scenes at corpus scale.
3. **Diagnostic**. Neighbourhood instability identifies scenes where joint scene-event models may struggle.

SCENE LABEL EXTRACTION PIPELINE



Acoustic scene corpus



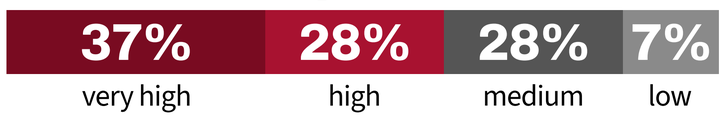
Identifiable scenes
- **130k** single-label
- **12k** multi-label
- **31%** unknown

Two taxonomies

186 scenes — **Audiovisual** · **12** roots

75 scenes — **Audio-centric** · **9** roots

Label reliability (model consensus)



Two ways to describe a scene

ACOUSTIC PROFILE

What sounds occur most frequently?

Computed from the corpus-wide scene-event co-occurrence matrix:

$$p(e \mid s) = \frac{c(e,s)}{\sum_{e'} c(e',s)}$$

*Dominated by **ubiquitous background sounds**.*

SEMANTIC PROFILE

What sounds characterize the scene?

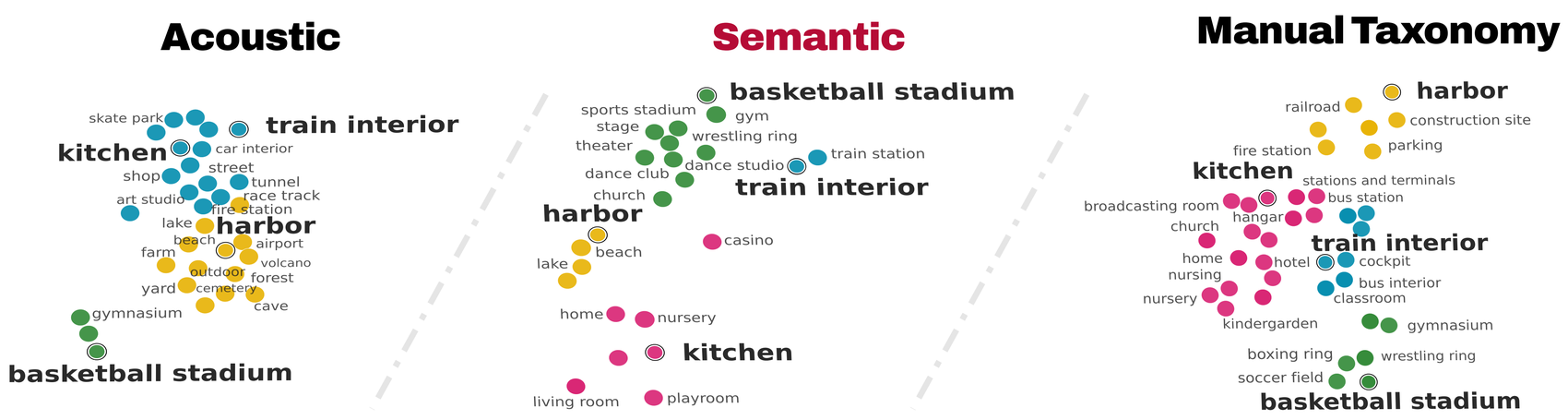
$$\mathrm{PMI}(e,s) = \log \frac{p(e \mid s)}{P(e)}$$

$$\mathrm{dPPMI}(e,s) = w(e,s) \cdot \max\left(0,\ \mathrm{PMI}(e,s)\right)$$

*Down-weights **shared sounds** and highlights **scene-specific sounds**.*

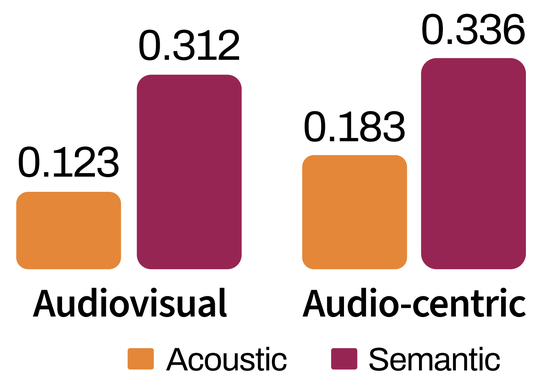
SCENES ANALYSIS

Representational structure of scenes



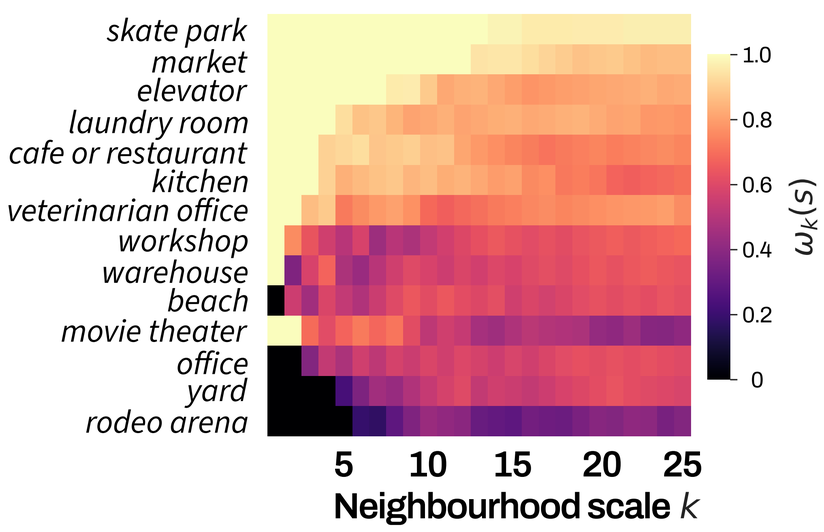
Similarity with human categorization

Spearman correlations with manual taxonomy



Where acoustics and semantics diverge

Acoustic - semantic shift



Neighbourhood instability $\omega_k(s) = 1 - \mathrm{RBO}_k(s)$, higher means the acoustic and semantic neighbours of a scene disagree.

Acoustic chameleons

High ω_k — **Scenes with shared background sounds**

skate park · market · kitchen · elevator

Semantic anchors

Low ω_k — **Scenes with aligned acoustic and semantic neighbours**

rodeo arena · office · yard · beach

Scene classification

Naive Bayes (NB) scene classifier from event labels, Macro-F1.

E.g., human voice · conversation · laughter → living room

Macro-F1	NB Acoustic	NB Semantic
Audiovisual	0.1282	**0.1315**
Audio-centric	0.1847	**0.1860**

NB Semantic improves due to gains for confusable scenes.

CONCLUSION

1. *We extended AudioSet Strong with **scene annotations** for **joint scene-event analysis**.*
2. ***Event semantics** link more strongly to **scene identity** than raw event frequency.*
3. ***Neighbourhood instability** signals where joint scene–event models may struggle.*

PROJECT WEBSITE: https://github.com/michelolzam/scenes-audioset

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